XENON INTERPRETABILITY

# Synthetic Market Phase 14

Affordance-ladder geometry on the same 4-asset market frame used in Phases 9-13. This phase asks whether progressively stronger execution constraints deform the shared market coordinates like risk and portfolio, or whether affordance behaves more like route masking.

| DATE | BASE DATA | STATES | OBJECT |
|---|---|---|---|
| 22 March 2026 | Phase 14 affordance ladder | `row_mean @ L1`, `row_eos @ L25` | adjacent affordance-step transforms |

**MAIN READ**

The shared 4-asset market frame survives the full affordance ladder almost perfectly early, but late states look sharper and more route-masking than either risk or portfolio.

Early `row_mean` states keep both latent axes above `0.990 R²` all the way to `affordance_5`, so the base market coordinates are still present. Late `row_eos` states do move toward affordance-adjusted score geometry, but the best adjacent fits favor flexible local maps and the full `market_only -> affordance_5` warp is only moderate. The strongest interpretation is preserved shared geometry plus increasingly strong route masking.

| EARLY FRAME FLOOR | LATE STRONGEST LOCAL FIT | LATE END-TO-END | NONTRIVIAL COMPOSE |
|---|---|---|---|
| **0.990 $R^2$** | **0.881 $R^2$** | **0.426 $R^2$** | **0.924 cos** |
| `market_only -> affordance_5`, latent_y | linear fit on `affordance_2 -> affordance_3` | best direct family on `market_only -> affordance_5` | late linear composed-vs-direct matrix cosine |

## Why Phase 14 Exists

Phases 12 and 13 established two complementary context stories on the same 4-asset object:

- risk preserved a shared market frame and behaved like a globally coherent near-rigid ladder
- portfolio preserved the same frame, but looked more local and redistributive

Affordance is the next useful family because it changes *execution availability* rather than preference weight. In the synthetic prompts:

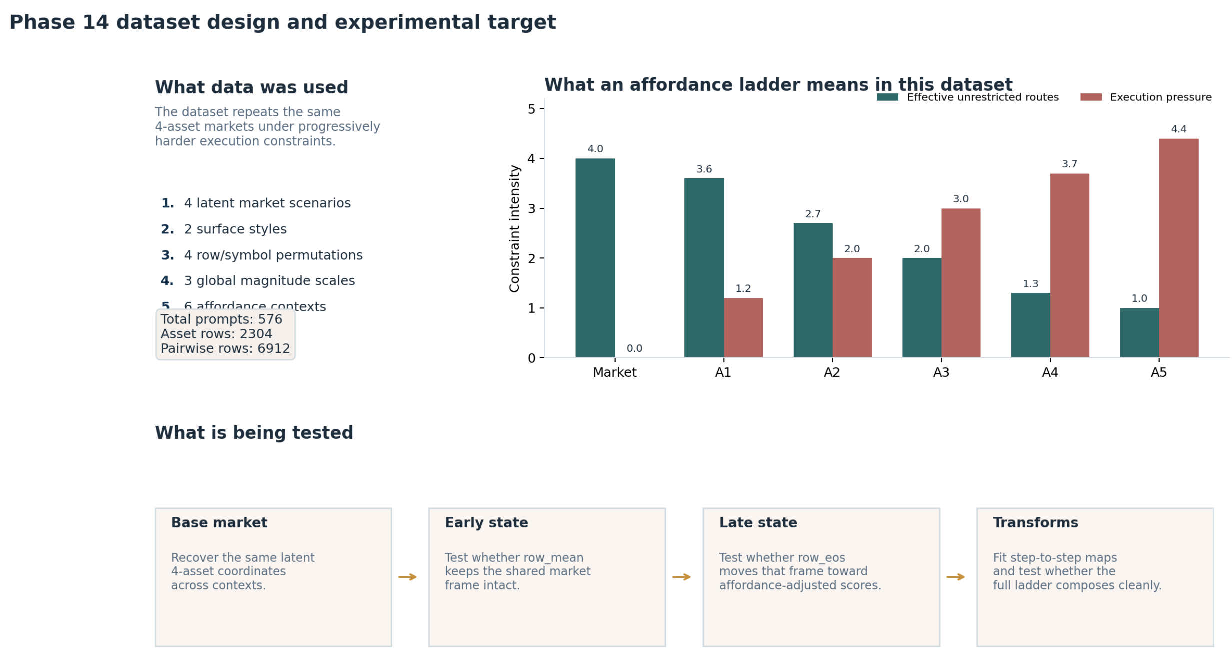
- the strongest asset is capped first
- the second asset becomes partially restricted and then blocked

- the third asset becomes size-limited late
- by `affordance_5`, only the weakest route remains fully open

So this phase asks whether the model treats execution constraints as:

- another smooth deformation of the shared market frame
- or a sharper route-masking process layered on top of that frame

## Experimental Design




Phase 14 is built from repeated variants of the same 4-asset markets. The figure shows the dataset recipe, what the affordance ladder means, and the actual question being tested.

The easiest way to read this phase is:

- start with a base 4-asset market
- render many nuisance variants of the same market
- place each variant into six contexts: `market_only`, then `affordance_1..affordance_5`
- ask whether the model keeps the same market coordinates and how those coordinates change across the ladder

Concretely, the dataset contains:

- 4 latent market scenarios
- 2 surface styles
- 4 row / symbol permutations
- 3 global scale variants
- 6 contexts per base market

That yields:

- `576` prompts total
- `2,304` asset rows
- `6,912` pairwise asset comparisons

In this report, the surface styles, row permutations, and global magnitude scales are all *nuisance variants*. That means they change how the same market is written on the page without changing the underlying latent market shape we actually care about.

## What "Affordance Ladder" Means

In this report, a "ladder" just means an ordered sequence of matched prompt variants. The market stays the same, but the contextual pressure changes step by step.

For Phase 14:

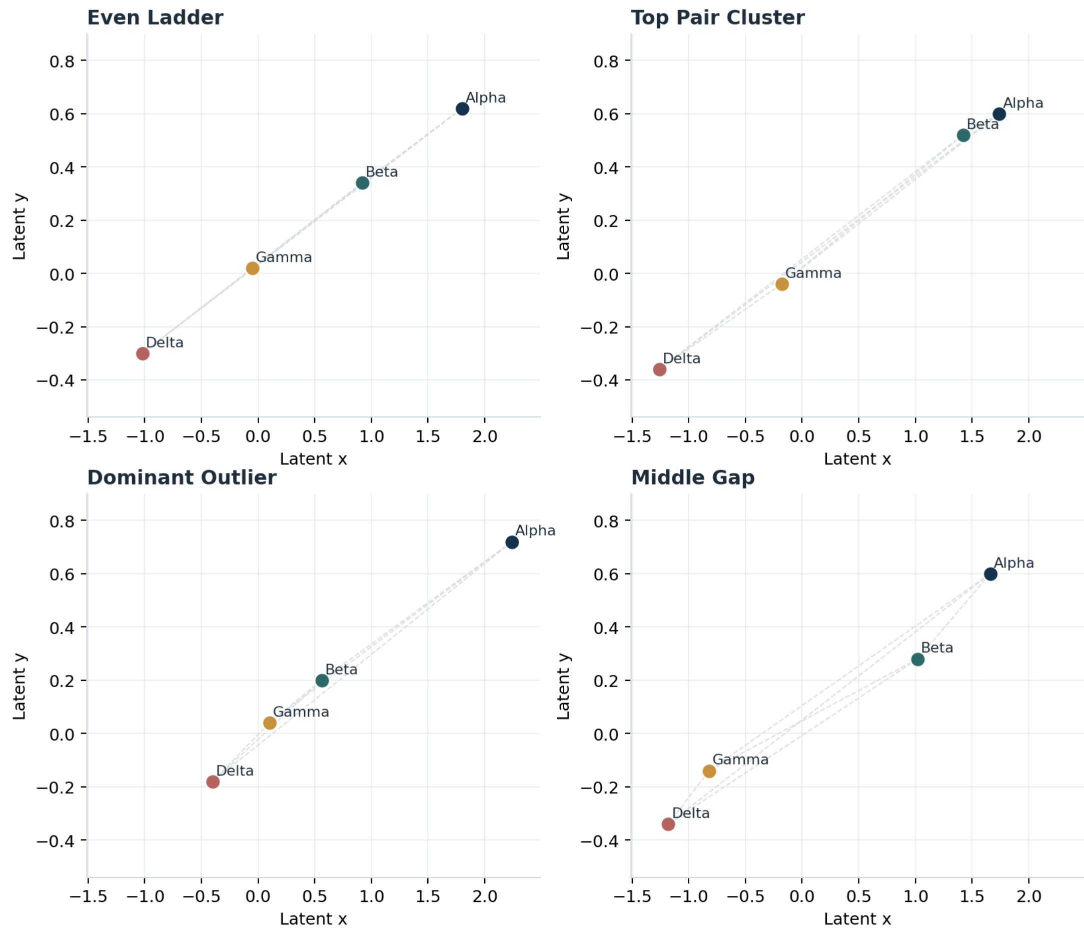
- `market_only` has no hard execution constraints
- `affordance_1` lightly caps the current leader
- `affordance_3` blocks the top two routes
- `affordance_5` leaves only the weakest-ranked route fully unrestricted

So the ladder is not a time series. It is a controlled constraint sweep over the same base market.

| Context | Constraint severity | Instructional effect |
|---|---|---|
| `market_only` | `none` | No hard execution constraints are supplied. |
| `affordance_1` | `light` | The strongest route is capped for new adds. |
| `affordance_2` | `light-medium` | The strongest route is capped and the next route becomes confirmation-only. |
| `affordance_3` | `medium` | The top two routes face hard execution limits. |
| `affordance_4` | `strong` | Top two blocked; third route can only be added in small size. |
| `affordance_5` | `strongest` | Only the weakest-ranked route remains broadly open. |

## What “4-Asset Market Geometry” Means

**The 4-asset geometry object: same assets, different whole-market shapes**




Each panel is a different latent market shape. The goal is not just to recover a winner, but to preserve the relative placement of all four assets at once.

The four latent market-shape families used in the dataset. Each point is one asset. Dashed edges mark the pairwise distances that collectively define the whole-market geometry.

The key object here is not a single asset row and not a single pairwise comparison. It is the *shape of the 4-asset market*.

That means:

- four assets are placed at known latent coordinates
- those coordinates define six pairwise distances
- the distances together determine the market’s overall shape

Why this matters:

- two markets can have the same winner but different shapes
- two markets can have the same rank order but different spacing or clustering
- a real set-level representation should preserve more than “which token looks best”

So when the report says "shared market frame," it means a reusable coordinate system that organizes all four assets together.

| Scenario name | Plain-language meaning |
|---|---|
| `even ladder` | The four assets step down in fairly regular intervals. There is a clear ordering, but no special clustering. |
| `top pair cluster` | The top two assets sit close together while the lower two are farther away. |
| `dominant outlier` | One asset is clearly separated above the other three, which are comparatively compressed. |
| `middle gap` | There is a larger-than-usual split between the upper pair and lower pair of assets. |

## Plain-Language Terminology

This is the shortest way to decode the jargon used later in the report:

- market frame the shared coordinate system the model seems to use to organize all four assets
- market coordinates the x/y position of each individual asset inside that shared frame
- market geometry the whole shape formed by those four coordinates together
- rigid map a transform that mostly keeps distances and angles intact, like a rotation or reflection
- context-adjusted score geometry the asset layout implied after we rescore the same market under a specific context, such as progressively stronger execution constraints
- nuisance variants prompt differences that should not change the underlying market meaning, like row order, symbol aliases, or formatting style

The point of Phase 14 is to separate:

- real geometry changes caused by context
- from fake changes caused by nuisance presentation differences

## How To Read The Layerwise Charts

Several figures in this report use the same general format: some metric is plotted *by layer*. The point of those charts is to show *where in the model* a representation or deformation is strongest.

The basic reading rules are:

- x-axis = layer moving left to right means moving from earlier model layers to later ones
- y-axis = average metric value higher usually means the tested representation is clearer or the tested context effect is stronger
- a peak the layer where that metric is strongest under the current comparison
- `row_mean` vs `row_eos` `row_mean` averages all tokens in a market row, while `row_eos` uses the row-ending token only

In this report, the two most common y-axis metrics are:

- mean Spearman a rank-correlation measure. Use this when the chart asks whether distances or relative orderings line up. `1.0` means the activation-space ordering matches the target ordering almost perfectly, `0` means little relationship, and negative values mean the ordering is inverted.
- margin a difference score between two candidate explanations. Here it usually means "how much closer is the recovered geometry to the context-adjusted score geometry than to the raw latent geometry?" Positive is evidence for realignment or deformation in the expected direction; values near `0` mean little separation; negative means the comparison is going the wrong way.

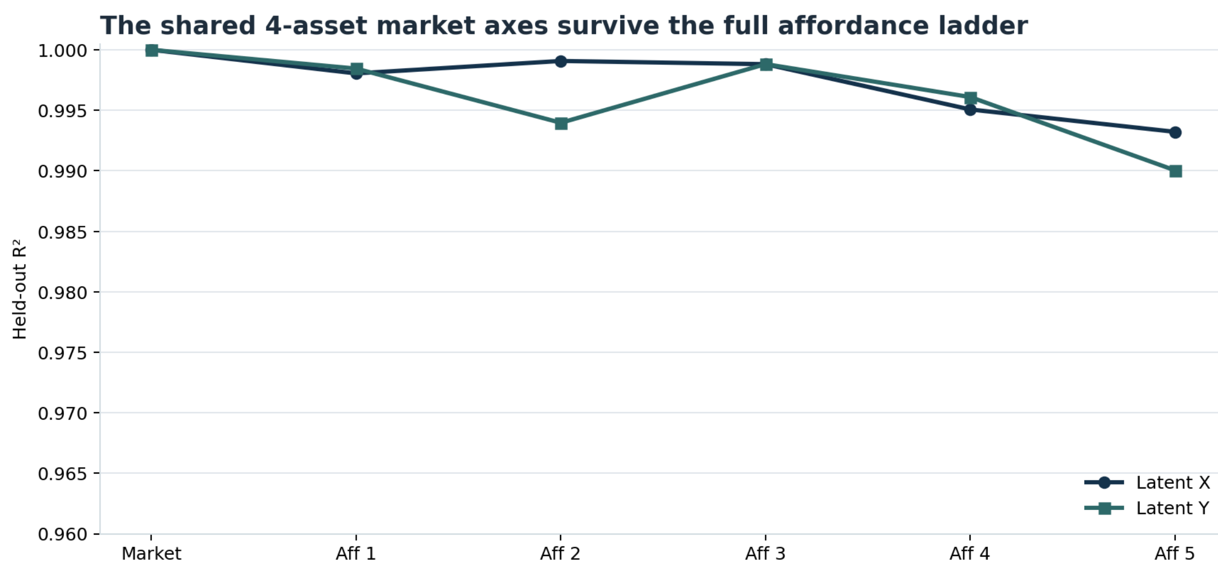
So a good mental model is:

- *high early peak* = the representation is present very early and may be close to raw market parsing
- *high late peak* = the effect emerges after more integration with context
- *broad plateau* = the effect is distributed across many layers
- *sharp isolated spike* = the effect is localized to a narrower part of the network

The figures below use those curves to answer two different questions:

- coordinate transfer asks whether the same base market frame survives across contexts
- realignment or deformation asks whether later layers actively move that frame toward a new context-specific geometry

## The Shared Market Frame Still Survives the Full Ladder




Held-out coordinate transfer from `market_only` into each affordance context. Both latent axes stay high if the shared frame survives.

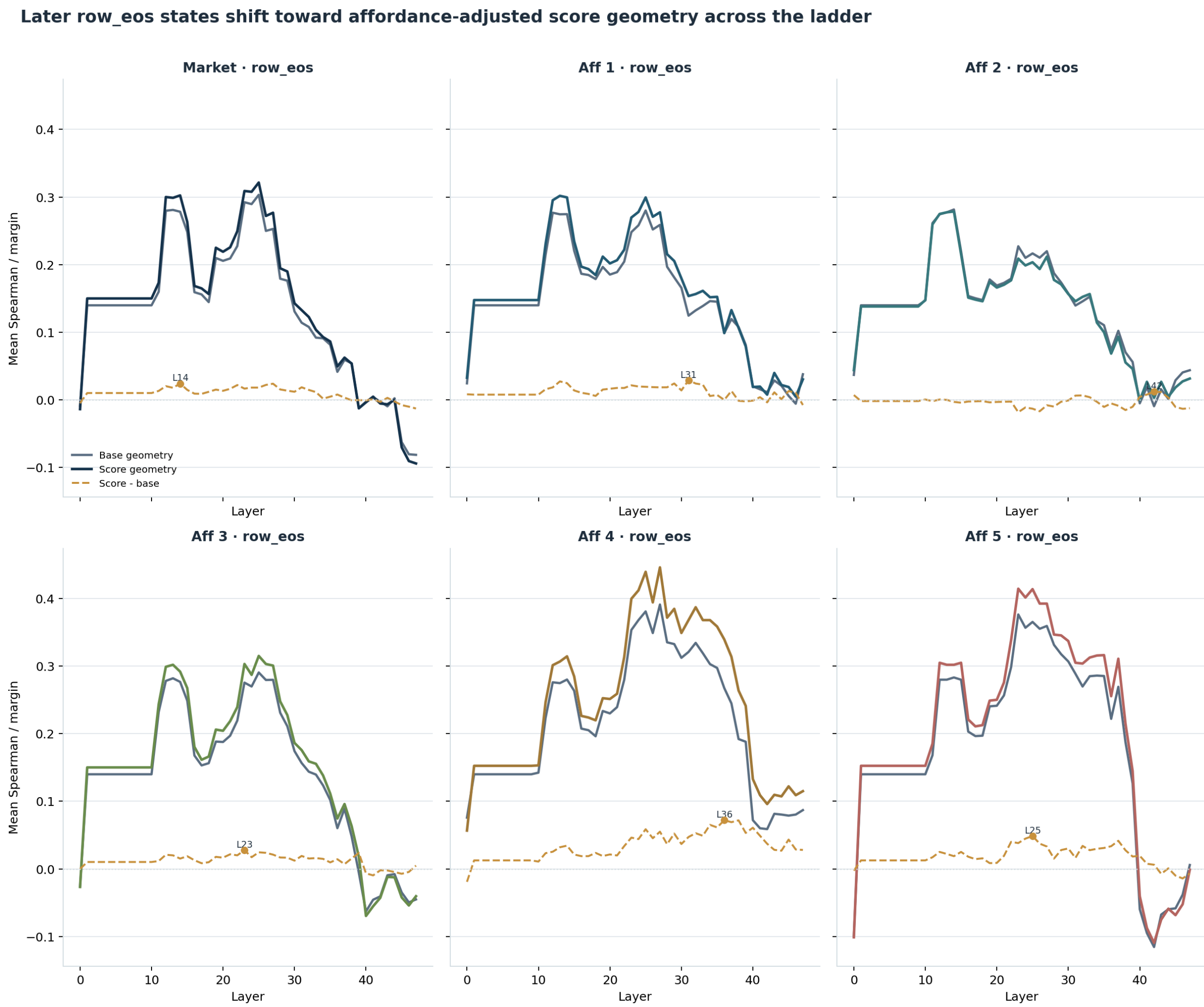
This is the strongest stability result in the report. A probe trained on `market_only` coordinates still recovers the same latent axes in the affordance contexts with very little loss:

- `market_only -> affordance_1`: `0.998` / `0.998` $R^2$ on `x` / `y`
- `market_only -> affordance_3`: `0.999` / `0.999` $R^2$
- `market_only -> affordance_5`: `0.993` / `0.990` $R^2$

The best layers remain extremely early: `row_mean @ L1` for most transfers, with only the `y` axis in `affordance_4` and `affordance_5` shifting slightly to `L4`.

So affordance does *not* erase the base market organization. Even when the strongest routes are progressively capped or blocked, the model still keeps a reusable 4-asset coordinate system that looks almost identical to the original market frame.

## Late States Still Realign Toward Affordance-Adjusted Score Geometry




Realignment margin between recovered activation-space geometry and affordance-adjusted score geometry for each context.

Late states do not stay purely latent. The best realignment margins all occur on `row_eos`, which means the end-of-row state is moving toward the affordance-adjusted score layout rather than staying at the raw latent market geometry.

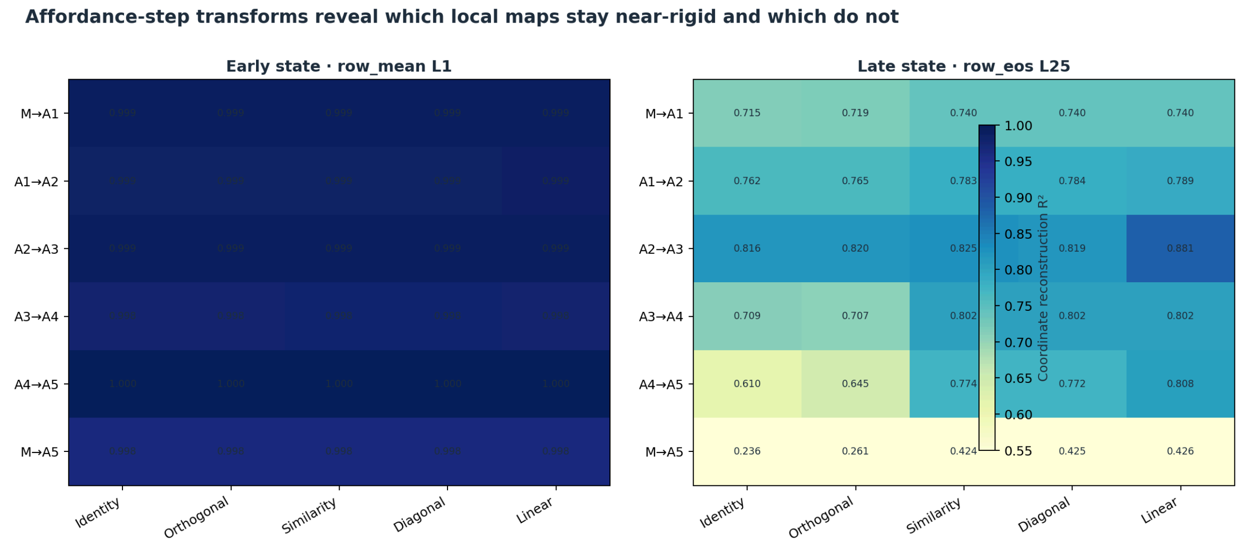
The pattern is uneven:

- `market_only`: margin `+0.024` at `L14`
- `affordance_1`: `+0.029` at `L31`
- `affordance_2`: only `+0.012` at `L42`

- `affordance_4`: strongest margin, `+0.072` at `L36`
- `affordance_5`: still strong, `+0.049` at `L25`

So the affordance ladder does reshape the late geometry, but it does so most clearly once the restrictions become genuinely severe. Mild execution caps are comparatively weak.

## Early Affordance Transforms Are Nearly Trivial



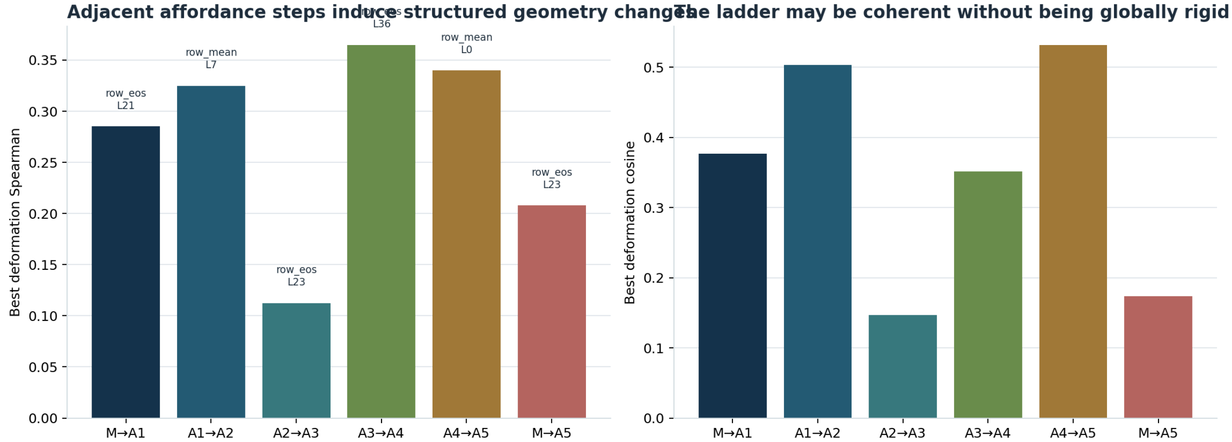

Coordinate reconstruction R² for transform families fit on each adjacent affordance step. Left: early state. Right: late state.

The early state is almost too clean. Every adjacent affordance step is already extremely well described by a trivial family:

- `market_only -> affordance_1`: best `diagonal`, `0.9993 R²`
- `affordance_1 -> affordance_2`: best `linear`, `0.9989 R²`
- `affordance_2 -> affordance_3`: best `linear`, `0.9994 R²`
- `affordance_3 -> affordance_4`: best `similarity`, `0.9984 R²`
- `affordance_4 -> affordance_5`: best `linear`, `0.9998 R²`
- full `market_only -> affordance_5`: best `identity`, `0.9976 R²`

That means the early recovered frame barely changes at all. The model is still carrying an almost unchanged market layout before the later context-sensitive reshaping begins.

# The Late Ladder Is Local, Mixed, or Mask-Like




Best deformation margins for adjacent and end-to-end affordance comparisons. Higher margins indicate stronger evidence that the geometry is shifting toward the context-adjusted arrangement.

This is where affordance starts to separate from risk and portfolio.

On the late state, the strongest adjacent-step fits are all local and flexible:

- `affordance_1 -> affordance_2`: best `linear`, `0.789` $R^2$
- `affordance_2 -> affordance_3`: best `linear`, `0.881` $R^2$
- `affordance_3 -> affordance_4`: best `linear`, `0.802` $R^2$
- `affordance_4 -> affordance_5`: best `linear`, `0.808` $R^2$

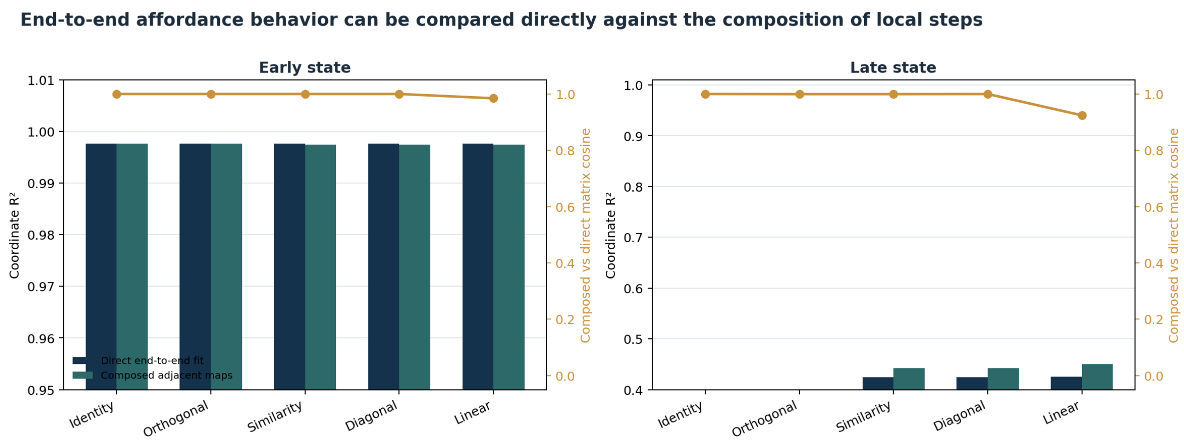
The deformation margins tell the same story. The largest positive shifts are not end-to-end:

- `affordance_3 -> affordance_4`: `+0.365`
- `affordance_4 -> affordance_5`: `+0.340`
- `affordance_1 -> affordance_2`: `+0.325`

By contrast, the direct end-to-end `market_only -> affordance_5` deformation is only `+0.208`, and its best direct transform fit is just `0.426` $R^2$.

So late affordance behavior looks less like one smooth global warp and more like a sequence of sharper local reallocations as routes get partially blocked and then removed.

## Composition Keeps Orientation Coherent, or Breaks It




For `market_only -> affordance_5`, compare direct end-to-end fits against the composition of all adjacent affordance-step maps. Gold marks matrix cosine between the composed and direct transforms.

The composition result is mixed in a revealing way.

For the late `market_only -> affordance_5` comparison:

- direct `identity`: `0.236 R²`
- direct `orthogonal`: `0.261 R²`
- direct `similarity`: `0.424 R²`
- direct `diagonal`: `0.425 R²`
- direct `linear`: `0.426 R²`

If we compose the adjacent-step maps instead of fitting the end-to-end step directly, the flexible families improve slightly:

- composed `similarity`: `0.443 R²`
- composed `diagonal`: `0.442 R²`
- composed `linear`: `0.451 R²`

But the matrix-cosine story is still constrained:

- `orthogonal`: `0.9994`
- `similarity`: `0.9994`
- `diagonal`: `0.9999`
- `linear`: `0.9242`

So the ladder keeps a broadly coherent orientation, but the late end-to-end geometry is not captured by a single clean rigid map. The best affordance story is a preserved base frame with increasingly strong local route-masking deformations layered on top.

## Interpretation

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> Affordance preserves the same shared 4-asset market frame very strongly in early states. Late states do move toward context-adjusted score geometry, especially once the strongest routes are heavily constrained. The best late adjacent-step fits prefer flexible `linear` transforms, which is consistent with local route closures or selective masking rather than simple preference reweighting.

> **NOT SUPPORTED**
>
> The full affordance ladder is not a single smooth global rigid map. Unlike risk, the late `market_only -> affordance_5` step is only moderately recoverable under the best family, and unlike the cleanest risk result, a single end-to-end orthogonal or identity map is not enough to explain the late geometry.

Putting Phases 11-14 together now gives a cleaner taxonomy:

- risk behaves like a globally coherent near-rigid ladder
- portfolio behaves like local redistributive movement inside the same frame
- affordance behaves like preserved frame plus sharper local route masking

That is a stronger and more interpretable result than a generic "context matters" claim. Different context families appear to deform the same underlying market coordinates in qualitatively different ways.

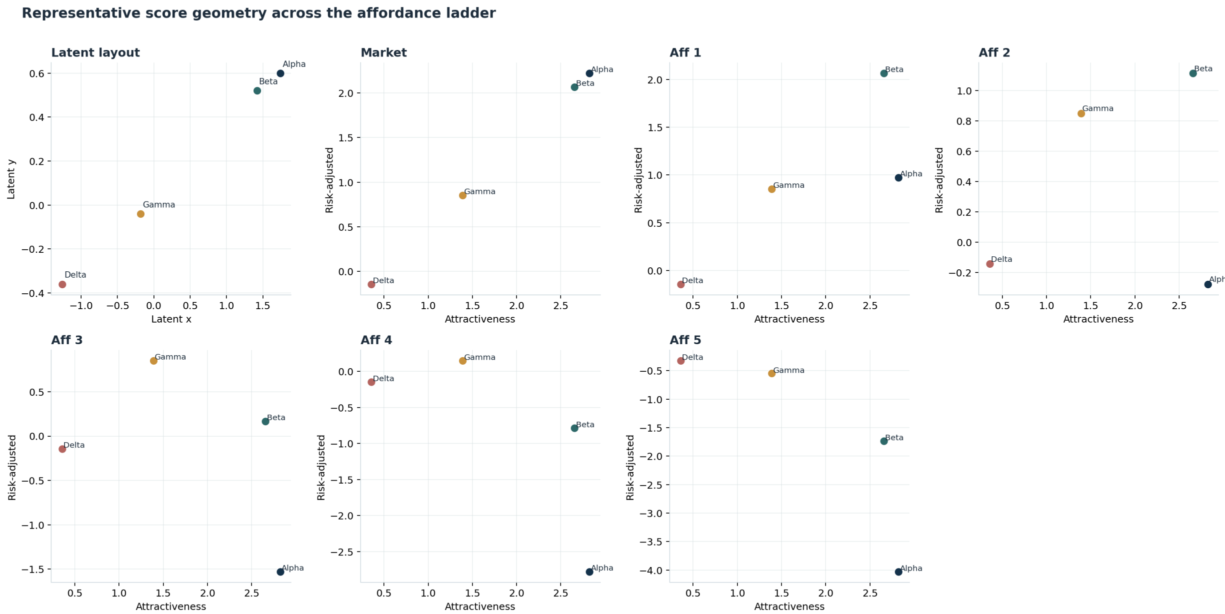
## What To Do Next

If Phase 14 keeps the shared frame but looks sharper and less globally coherent than both risk and portfolio, the next step should be a small real-DX bridge using matched prompt variants and the same coordinate-and-deformation lens.

The key question is no longer "does a latent space exist?" The key question is:

- do real execution overlays also look like structured deformations of a shared market frame
- and if so, which context families are smooth, local, or mask-like

# Appendix A: Representative Affordance Geometry




Representative `top_pair_cluster` market from the affordance ladder. The figure shows the same latent base market with affordance-adjusted score geometries layered on top of it.

This figure anchors what "affordance deformation" means in the report. The transform analyses above are attempts to recover how the same underlying 4-asset market is reweighted once the strongest execution paths are progressively capped or blocked.

# Appendix B: Glossary

| Term | Meaning |
|---|---|
| `shared market frame` | The recovered 2D coordinate system that organizes all four assets before context-specific reweighting. |
| `market coordinates` | The individual x/y positions of the four assets inside the shared market frame. |
| `market geometry` | The whole-market arrangement of those coordinates, including spacing, clustering, and pairwise distances. |
| `4-asset geometry` | The full relative placement of all four assets at once. In practice this means the coordinate layout or, equivalently, the pattern of six pairwise distances between assets. |
| `affordance ladder` | A matched sequence of context variants where the base market is fixed and execution availability is stepped from `market_only` to `affordance_5`. |
| `nuisance variants` | Prompt changes that alter presentation but should not alter the underlying market meaning, such as row order, formatting style, symbol aliases, or global scale formatting. |

| Term | Meaning |
|---|---|
| `coordinate transfer` | How well a probe trained in one context can recover the same latent coordinate in another context. High $R^2$ means the base frame survived. |
| `realignment margin` | How much closer the recovered geometry is to the context-adjusted score geometry than to the raw latent layout. Positive margins mean the context is actively reweighting the market. |
| `context-adjusted score geometry` | The geometry implied by rescoring the same four assets under the current context. In Phase 14, that means incorporating progressively stronger execution caps and blocks. |
| `identity` | No transform at all. If identity already works, the two contexts share almost the same geometry in the decoded frame. |
| `orthogonal` | A pure rotation or reflection. Distances and angles are preserved; only orientation changes. |
| `rigid map` | A transform that preserves shape up to orientation. In practice here, the closest approximation is an orthogonal map or a near-rigid similarity map. |
| `similarity` | Uniform scaling plus rotation. This captures isotropic compression or expansion. |
| `diagonal` | Axis-aligned rescaling in the shared coordinate frame. This captures simple non-uniform compression without rotation. |
| `linear` | Any 2×2 linear map. This is the most flexible family here and can absorb rotation, anisotropic scaling, and shear-like behavior. |
| `matrix cosine` | Cosine similarity between two flattened transform matrices. Here it is used to compare direct end-to-end maps against the composition of adjacent ladder steps. |